MODULE 4 – HADOOP ECOSYSTEM

Learning Outcomes

By the end of this module, students should be able to:

- Explain the major components of the Hadoop ecosystem.
- Differentiate between Hadoop's storage (HDFS), resource management (YARN), and processing (MapReduce, Hive, Pig, etc.).
- Run basic MapReduce jobs.
- Understand how Hive and Pig simplify working with large datasets.
- Integrate Hadoop ecosystem tools into end-to-end workflows.

Theory Overview

The Hadoop ecosystem consists of multiple open-source projects designed to support large-scale data storage and processing.

Core Components

- **HDFS (Hadoop Distributed File System):** Distributed, fault-tolerant storage.
- **YARN (Yet Another Resource Negotiator):** Manages resources and schedules jobs across the cluster.
- **MapReduce:** Programming model for large-scale batch processing.

Data Processing & Querying Tools

- **Apache Hive:** Data warehouse tool; allows SQL-like queries (HiveQL) on HDFS data.
- **Apache Pig:** High-level dataflow language (Pig Latin) for ETL and transformations.
- **Apache HBase:** Column-oriented NoSQL database built on top of HDFS.

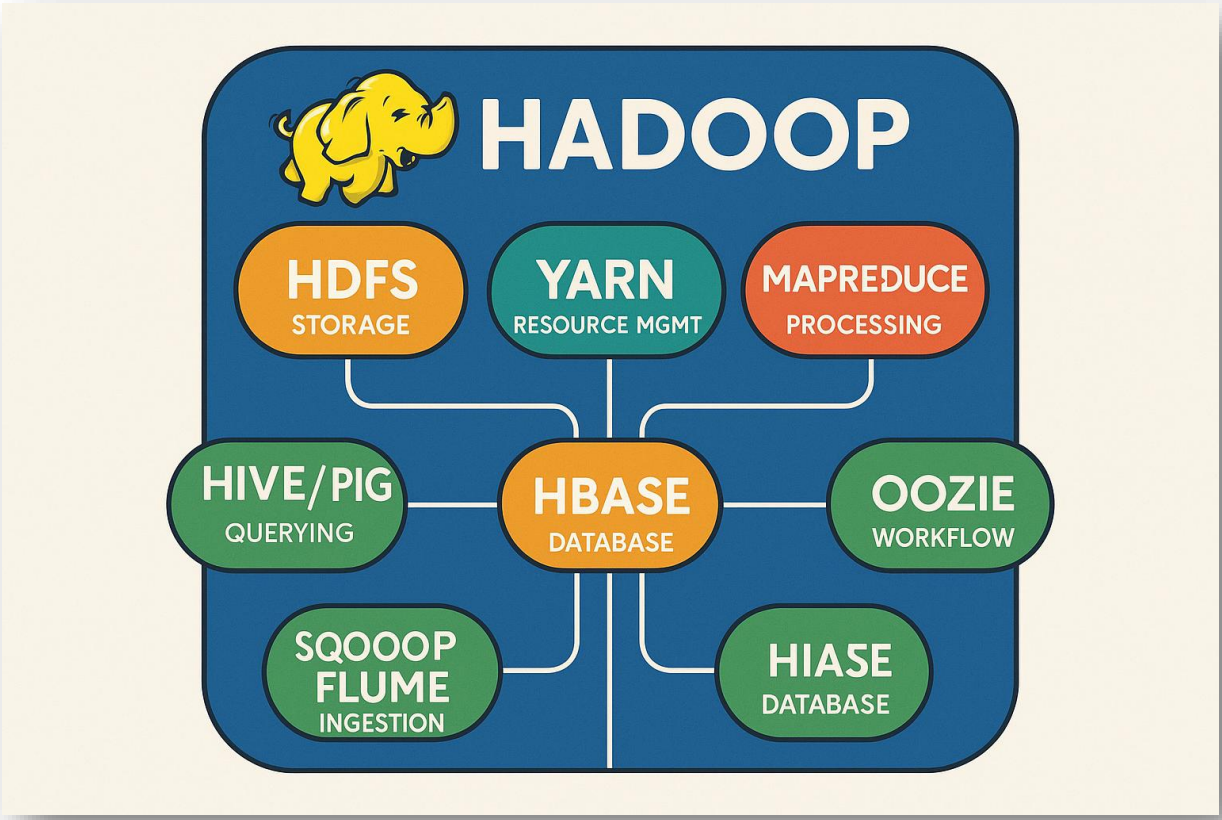
Data Ingestion & Integration

- **Apache Sqoop:** Transfers data between relational databases (MySQL, PostgreSQL, Oracle) and Hadoop.
- **Apache Flume:** Collects, aggregates, and moves large amounts of log data into HDFS.

Workflow & Orchestration

- **Apache Oozie:** Job scheduling and orchestration for Hadoop workflows.

Hadoop Ecosystem Components



Lab Exercise — Running Hadoop and Hive

Objective:
Run a simple MapReduce word count job and query data using Hive.

Requirements

- Hadoop (pseudo-distributed or cluster mode).
- Hive installed and configured.

Step-by-Step Tasks

```
# 1. Place input text into HDFS (bash)
hdfs dfs -put sample.txt /user/student/input/

# 2. Run built-in MapReduce word count example (bash)
hadoop jar /usr/local/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /user/student/input /user/student/output

# 3. View results (bash)
hdfs dfs -cat /user/student/output/part-r-00000 | head

# 4. Create Hive table from a CSV file in HDFS (sql)
CREATE TABLE students (
    id INT,
    name STRING,
    gpa FLOAT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE;

# 5. Load data and run a query (sql)
LOAD DATA INPATH '/user/student/students.csv' INTO TABLE students;
SELECT name, gpa FROM students WHERE gpa > 3.0;
```

Sample Code — MapReduce in Python (Hadoop Streaming)

```
# mapper.py
#!/usr/bin/env python3
import sys
for line in sys.stdin:
    for word in line.strip().split():
        print(f"{word}\t1")

# reducer.py
#!/usr/bin/env python3
import sys
from collections import defaultdict

counts = defaultdict(int)
for line in sys.stdin:
    word, num = line.strip().split("\t")
    counts[word] += int(num)

for word, count in counts.items():
    print(f"{word}\t{count}")

# Run with Hadoop Streaming (bash)
hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-*.jar   -input /user/student/input   -output /user/student/output2   -mapper mapper.py   -reducer reducer.py
```

Case Study:

A national immunization registry integrates childhood vaccination data across provinces. Researchers need insights into coverage rates and dropout patterns.

Big Data Approach:

- Use Hive to run SQL queries over millions of vaccination records.
- Apply Pig scripts for ETL (extract birth cohorts, filter missed doses).
- Integrate with HBase for quick lookups on individual immunization histories.

Expected Outcomes:

Detect under-vaccinated districts, predict outbreaks of preventable diseases, and inform public health interventions.

Suggested Readings

- Tom White, *Hadoop: The Definitive Guide*.
- Alan Gates, *Programming Pig*.
- Dean Wampler et al., *Programming Hive*.
- Apache HBase official documentation.